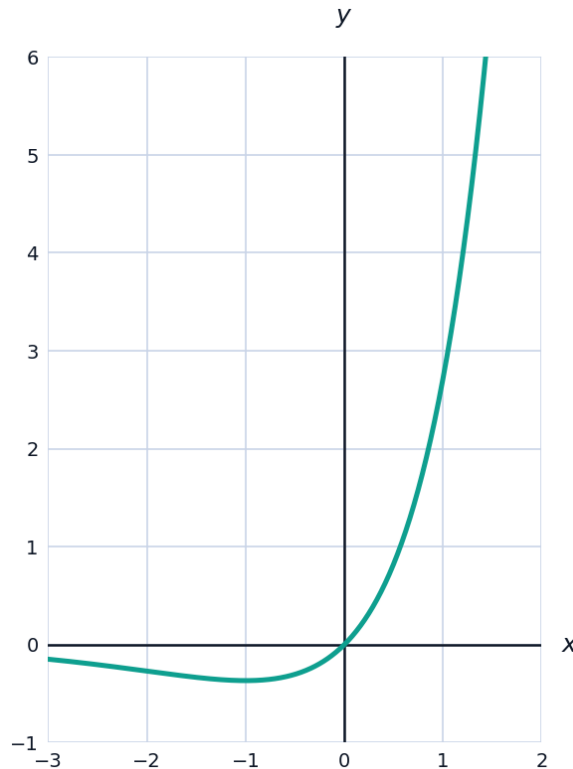


Integration by Parts



The integration by parts formula

- 1 State the integration by parts formula, and explain the LIATE guideline for choosing u .
- 2 Derive the integration by parts formula starting from the product rule $\frac{d}{dx}[uv] = u'v + uv'$.
- 3 Evaluate $\int xe^x dx$.



- 4 Evaluate $\int x \cos x dx$.

Integration by parts with logarithms and inverse trig

- 5 Evaluate $\int \ln x dx$.
- 6 Evaluate $\int x \ln x dx$.
- 7 Evaluate $\int \tan^{-1} x dx$.
- 8 Evaluate $\int x^2 \ln x dx$.

- 9 Evaluate $\int \sin^{-1} x \, dx$.
-

Repeated integration by parts

- 10 Evaluate $\int x^2 e^x \, dx$, applying integration by parts twice.
- 11 Evaluate $\int x^2 \sin x \, dx$, applying integration by parts twice.
- 12 Set up (do not evaluate fully) the reduction needed to find $\int x^3 e^x \, dx$, identifying how many times integration by parts must be applied.
-

The cyclic case

- 13 Evaluate $\int e^x \sin x \, dx$ using the cyclic (solve-for-the-integral) technique.
- 14 Evaluate $\int e^{2x} \cos x \, dx$ using the cyclic technique.
-

Definite integrals using integration by parts

- 15 Evaluate $\int_0^1 x e^x \, dx$ using the antiderivative found earlier.
- 16 Evaluate $\int_1^e \ln x \, dx$ using the antiderivative found earlier.
- 17 Evaluate $\int_0^\pi x \sin x \, dx$.
-

Integration by parts to derive a reduction formula

- 18 Use integration by parts to show $\int x^n e^x \, dx = x^n e^x - n \int x^{n-1} e^x \, dx$ (the reduction formula), and use it once to reconfirm $\int x^2 e^x \, dx$.
-

Choosing u and dv wisely

- 19 For each integral, state the best choice of u (using LIATE) without evaluating:
- | | |
|------------------------------|---------------------------|
| a $\int x^3 \ln x \, dx$ | b $\int x e^{5x} \, dx$ |
| c $\int x \tan^{-1} x \, dx$ | d $\int x^2 \cos x \, dx$ |

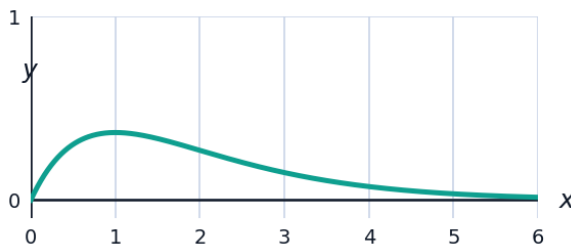
- 20 Explain what goes wrong if you attempt $\int x e^x dx$ with the reversed choice $u = e^x$, $dv = x dx$.
-

Integration by parts with algebraic-trigonometric products

- 21 Evaluate $\int x \sec^2 x dx$.
- 22 Evaluate $\int 2x \sin(3x) dx$.
-

Definite integrals: area under a transcendental product

- 23 Find the exact area under $y = x e^{-x}$ on $[0, 3]$, using $\int x e^{-x} dx = -x e^{-x} - e^{-x} + C$ (verify this antiderivative first by differentiating).



- 24 Evaluate $\int_0^{\pi/2} x \cos x dx$ using the antiderivative $x \sin x + \cos x$ found earlier.
-

Mixed review: choosing the right technique

- 25 For each integral, state whether it should be solved by u -substitution or integration by parts, and identify the key choice (substitution variable or u) without fully evaluating:
- | | |
|-----------------------------|---------------------|
| a $\int 2x e^{x^2} dx$ | b $\int x^2 e^x dx$ |
| c $\int \frac{\ln x}{x} dx$ | d $\int x \ln x dx$ |